

Writing Large Numbers in Scientific Notation*Question Bank - Additional 3.08*

Show your work. Circle your final answer.

For questions 1 through 4, write each number in scientific notation.

1. 70,000

2. 2,800,000

3. 640,000,000

4. 5,030,000,000

For questions 5 and 6, write each number in standard form.

5. 4.1×10^5

6. 1.64×10^7

Real quantities.

7. Write each quantity in scientific notation.

- a. Earth's beaches hold about 7,500,000,000,000,000 grains of sand.
- b. A distance of about 39,900,000,000,000 km.
- c. A computer displays a population as **1.2E7**. Write it in standard form.

8. Complete each comparison.

- a. How many times larger is 8.2 than 4.1?
- b. How many times larger is 10^7 than 10^5 ?
- c. How many times larger is 8.2×10^7 than 4.1×10^5 ?

9. Answer each question.

a. About 6.94×10^6 people visited one theme park in a year, and about 1.7×10^6 visited another. About how many times larger is the first number? Round to the nearest whole number.

b. China's population is about 1.44×10^9 and Mexico's is about 1.292×10^8 . About how many times larger is China's? Round to the nearest tenth.

10. Which of the following is greater? Explain how you know.

a. 8.7×10^4 or 2.1×10^5

b. 5.6×10^6 or 5.09×10^6

11. Which is 71,000,000 written in scientific notation?

- ☐ A. 7.1×10^8
- ☐ B. 71×10^6
- ☐ C. 7.1×10^6
- ☐ D. 7.1×10^7

12. Select ALL of the expressions that are equal in value to 6,200,000.

- ☐ A. 6.2×10^5
- ☐ B. 620×10^4
- ☐ C. 0.62×10^7
- ☐ D. 6.2×10^7
- ☐ E. 62×10^5
- ☐ F. 6.2×10^6

13. In question 12, only one of the four equal expressions is written in scientific notation. Which one is it, and why are the other three not?